Analysis for Lab Exercise: Exploring Neural Information Processing

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AI in Healthcare (ITAI 4375) Spring 2026

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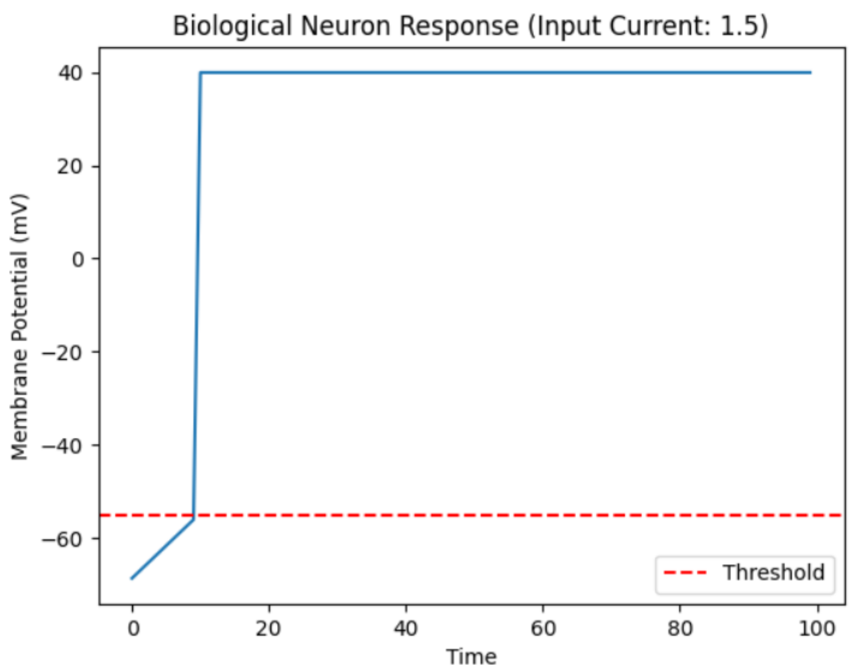
April 10, 2026

Stakeholder Engagement Plan for an AI Diagnostic Tool in Healthcare

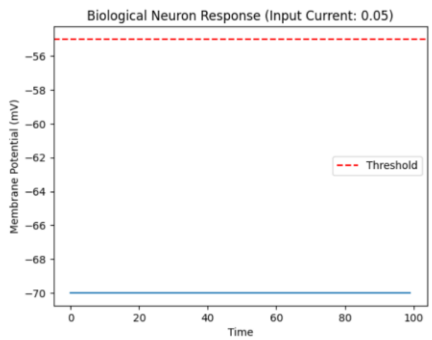
Introduction

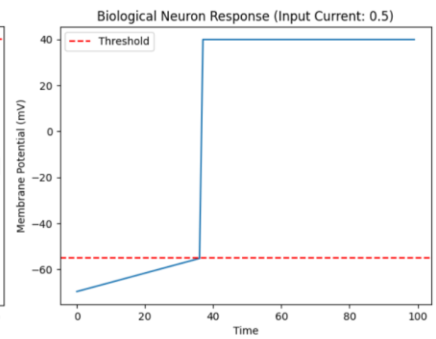
Biological and artificial neurons both process information, but they do so in very different ways. In this lab, I explored how a simplified biological neuron responds to input current, how an artificial neuron transforms numerical inputs through activation functions, and how a small feedforward network integrates information. These activities helped me better understand the connection between neuroscience and artificial intelligence. The lab showed that modern AI is inspired by the brain, but it is not a direct copy of biological neural processing. Instead, artificial neural networks take a few key ideas from biology and turn them into mathematical tools that are easier to compute at scale. The results of this lab made it clear that artificial neurons are powerful because of their simplicity, but they also leave out many important features of real neurons.



Graph for Step 2: Simulating a Biological Neuron

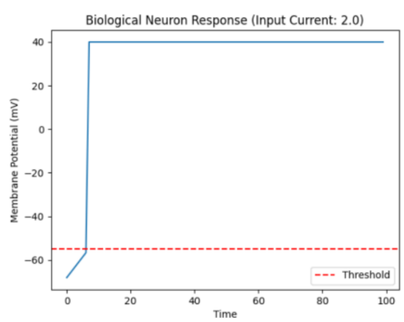
One of the most important concepts in the biological portion of the lab was the action potential. Biological neurons communicate through electrical and chemical signals, and in this assignment the focus was on the electrical side of that communication. A real neuron stays at a resting membrane potential until input raises its voltage toward a threshold. Once that threshold is crossed, the neuron fires an action potential. This behavior reflects the all-or-none principle, meaning the neuron does not produce a partial spike. It either fires or it does not. In the simplified biological model used in the lab, the membrane potential started at a resting value, increased when input current was applied, and generated a spike when it reached threshold. This model also included a simple decay mechanism to represent the return toward resting potential over time. Even though the code was simplified, it still captured the main idea that biological information processing is dynamic, time-based, and dependent on changes in membrane voltage.





Experiment 1: Sub-threshold Input

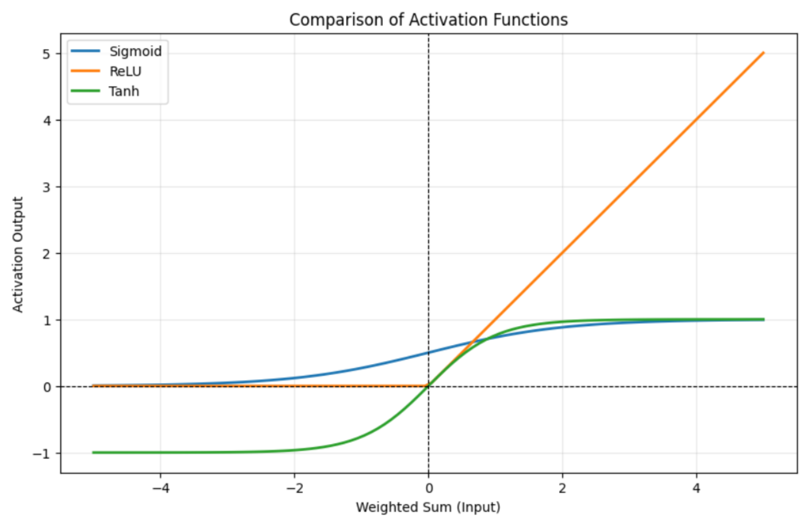
Experiment 2: Moderate Input Current



Experiment 3: Strong Input Current

The experiments with different input currents helped show how biological neurons respond to varying levels of stimulation. When the input current was very weak, the membrane potential rose only slightly and never reached threshold, so no action potential was produced. This demonstrated that sub-threshold stimulation does not lead to firing. When the input current was moderate, the membrane potential increased more steadily and eventually crossed threshold, producing a spike. When the input current was strong, the threshold was reached much faster, so the neuron fired more quickly. A key lesson from this observation is that stronger input does not necessarily make the spike itself larger. Instead, it changes how soon or how often the neuron fires. This is important because biological neurons often encode stimulus intensity through timing and firing frequency rather than by changing the amplitude of the spike. That finding highlights the fact that biological computation is strongly tied to time, accumulation, threshold crossing, and response patterns.

The artificial neuron model worked very differently. Instead of modeling membrane voltage over time, it took a set of numerical inputs, multiplied them by weights, added a bias term, and passed the result through an activation function. This means the artificial neuron does not naturally represent the continuous temporal behavior of a biological neuron. It does not accumulate voltage over time, model refractory periods, or simulate ionic processes. Instead, it performs a direct mathematical transformation from input to output. In the lab, this made it easier to observe how changes in weights or activation functions affect the final result. The artificial neuron was simpler, cleaner, and much more computationally efficient than the biological model. However, that efficiency comes from abstraction. Many of the complex processes that make biological neurons adaptive and realistic are removed in order to create a model that is easier to use in machine learning systems.



Graph for Exploring Activation Functions

The activation function portion of the lab further emphasized the difference between biological and artificial processing. The sigmoid, ReLU, and tanh functions each shaped the output of the artificial neuron in a different way. Sigmoid compressed values into a range between 0 and 1, which is useful when an output needs to be interpreted like a probability. Tanh mapped inputs into a range between -1 and 1, making it zero-centered and often more balanced for certain learning tasks. ReLU output zero for negative inputs and increased linearly for positive inputs, which introduced sparsity and has become especially useful in deep learning. These functions can be compared loosely to the threshold behavior of biological neurons, but the relationship is only partial. A biological threshold is a physical event that leads to an action potential, whereas an activation function is a mathematical rule that shapes output continuously or piecewise. ReLU is the closest simple analogy to a threshold, but even it is still only a rough abstraction of a far more complex biological process.

Comparing the two models shows several major differences in information processing. First, biological neurons are time-dependent, while artificial neurons are usually instantaneous. A biological neuron integrates information over time and produces a discrete event when threshold is reached. An artificial neuron typically takes a snapshot of its input and immediately computes an output. Second, biological neurons are physically and chemically complex, whereas artificial neurons are simplified mathematical objects. Real neurons contain ion channels, dendrites, axons, neurotransmitters, synapses, and refractory mechanisms. Artificial neurons reduce this complexity to weights, biases, and activation functions. Third, biological neurons communicate through spikes, while artificial neurons usually communicate through scalar numerical values. These are major differences, but there are also similarities. Both systems transform inputs into outputs, both can be connected into larger networks, and both rely on the interaction of many units rather than the power of a single isolated neuron. This similarity is one reason artificial neural networks remain such an important approach in AI.

The network portion of the lab demonstrated why collections of simple units can become useful for more advanced processing. A single neuron can only perform a limited transformation, but a small feedforward network can combine inputs, create intermediate representations, and produce more flexible outputs. In the experiment, different input patterns produced different hidden-layer activations and different final outputs. This showed that network architecture matters. The number of layers, the number of neurons, and the choice of activation functions all influence how information is processed. A hidden layer allows the system to transform raw inputs before making a final decision, which is why networks are more powerful than single neurons. This small experiment helped illustrate the core idea behind neural networks in AI: intelligence in these systems comes less from individual neurons and more from the way many simple units are connected and organized.

At the same time, the lab made it clear that simple artificial neuron models have important limitations. One limitation is that they oversimplify biology. Artificial neurons do not capture dendritic integration, spike timing, refractory periods, neurotransmitter effects, or the detailed electrochemical behavior of real neurons. Another limitation is the lack of built-in temporal memory in a basic feedforward model. Each input pattern is processed independently, which makes the system weak for tasks involving sequence, history, or ongoing context. A third limitation is the rigid architecture of simple artificial networks. Biological brains are highly plastic. Connections strengthen or weaken with experience, and neural systems are able to adapt continuously. Simple artificial networks, by contrast, usually rely on fixed structures that only change during training. In addition, small networks have limited capacity. With few neurons and parameters, they can only represent simple patterns and struggle with richer, more nonlinear problems.

These limitations point directly to ways biological principles could inspire more sophisticated and efficient AI systems. One area of inspiration is temporal processing. Since real neurons operate over time, future AI systems could improve by better modeling timing, sequence, and event-based computation. Another area is energy efficiency. The brain performs remarkable amounts of computation using far less energy than most modern AI systems. That suggests that sparse signaling, event-driven computation, and more efficient hardware could make AI more practical and sustainable. A third biological principle is plasticity. Real brains adapt continuously, and AI could become more flexible by using systems that learn more dynamically instead of depending entirely on static architectures and large retraining cycles. Finally, biological systems are robust and fault tolerant. Even when some neurons fail or signals are noisy, the brain continues functioning. AI systems inspired by this resilience could become more reliable in real-world environments where data are messy and conditions are unpredictable.

One especially promising direction is the development of more biologically plausible neural models, such as spiking neural networks. These models move closer to the event-based behavior of real neurons by representing information through spikes rather than only continuous activation values. While they are still far from fully capturing the brain, they reflect a growing recognition that time, sparsity, and biologically inspired computation may help push AI forward. Even if artificial intelligence never becomes an exact imitation of biology, neuroscience still provides a valuable framework for identifying what current AI systems are missing and where future improvements may be found. This lab reinforced that idea by showing that the gap between biological and artificial neurons is both a limitation and an opportunity.

In conclusion, this lab provided a useful comparison between biological and artificial information processing. The biological neuron model emphasized threshold behavior, time dependence, and action potentials, while the artificial neuron model emphasized weighted sums, activation functions, and computational efficiency. The network experiments showed that combining simple units can create more flexible behavior, but they also revealed how limited small feedforward systems remain when compared with real biological intelligence. Overall, the lab demonstrated that artificial neural networks are inspired by the brain but are much simpler than the systems they are modeled after. That simplicity is the source of both their power and their weakness. Artificial neurons are effective because they make computation manageable, but future AI systems may become even more advanced by borrowing additional ideas from biology, especially in areas such as temporal dynamics, plasticity, efficiency, and adaptive organization. This makes neuroscience not only a source of inspiration for AI, but also a guide for where the next generation of intelligent systems may go.